Enterprise Network Design Report

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1. Introduction

This report presents the design and implementation of a multi-site enterprise network; connecting London, Leeds, Liverpool, and Leicester. The network is structured using VLAN segmentation to separate departments (Finance, Executive, and Employees) and uses OSPF as a dynamic routing protocol to enable communication between sites. Core services such as DHCP are configured for automatic IP address allocation, while NAT is implemented on the London router to provide controlled internet access. Security is enforced using Access Control Lists (ACLs) to restrict inter-department communication and limit external access, ensuring a scalable, efficient, and secure network infrastructure.

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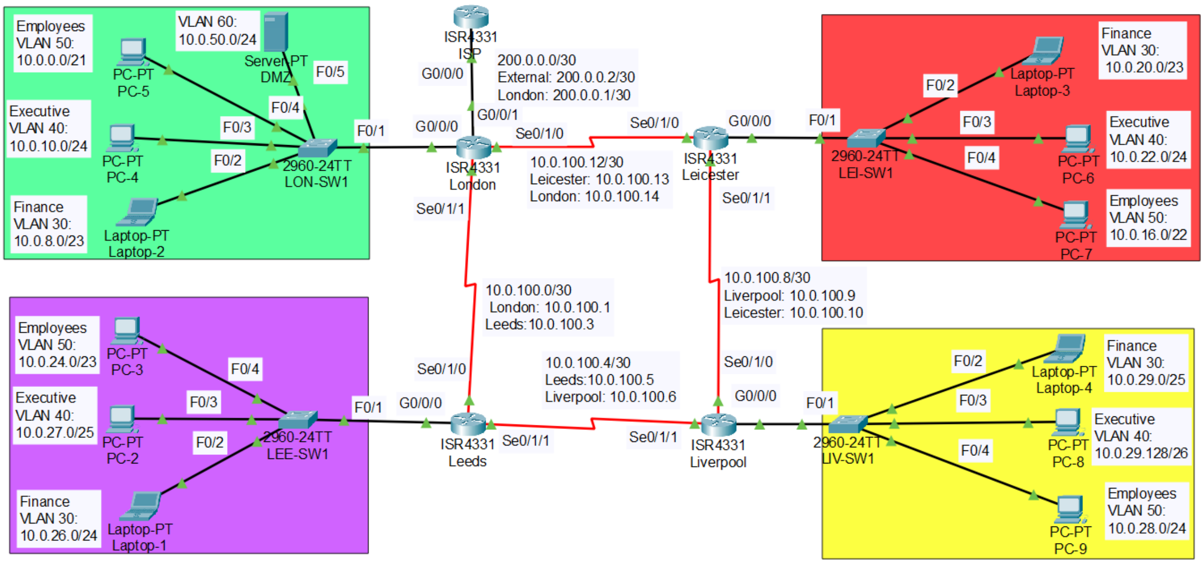
2. Network Design Overview

The network consists of four geographically distributed sites interconnected through serial links, forming a routed WAN environment. Each site includes a router and a switch configured with multiple VLANs to logically segment departments. London acts as the central hub, hosting the NAT configuration and connection to the simulated ISP network. The design follows a hierarchical approach, separating internal LAN traffic from external communication, and ensures redundancy and efficient routing through OSPF. This structure improves manageability, security, and scalability of the enterprise network.

2.1. Proposed Network Topology

- Implementation of four-site interconnected network architecture.
- Primary internet gateway at the London branch with controlled access for other branches.
- Redundant links between branches to ensure network reliability.



<<My Network Topology>>

2.2. VLAN Implementation

Three VLANs will be established per site:

- VLAN 30 - Finance (restricted access for financial transactions)
- VLAN 40 - Executive (secured for management use)
- VLAN 50 - Employees (general workforce operations)

Each VLAN will be managed via Layer 3 switches with inter-VLAN routing enabled.

London				
VLAN	**Purpose**	**Network Address**	**Subnet Mask**	**Hosts**
VLAN 50	Employees	10.0.0.0	255.255.248.0 (/21)	2046
VLAN 30	Finance	10.0.8.0	255.255.254 (/23)	510
VLAN 40	Executive	10.0.10.0	255.255.255.0 (/24)	254

Leicester				
VLAN	**Purpose**	**Network Address**	**Subnet Mask**	**Hosts**
VLAN 50	Employees	10.0.16.0	255.255.252.0 (/22)	1022

VLAN 30	Finance	10.0.20.0	255.255.254.0 (/23)	510
VLAN 40	Executive	10.0.22.0	255.255.255.0 (/24)	254

Liverpool				
VLAN	**Purpose**	**Network Address**	**Subnet Mask**	**Hosts**
VLAN 50	Employees	10.0.28.0	255.255.255.0 (/24)	254
VLAN 30	Finance	10.0.29.0	255.255.255.128 (/25)	126
VLAN 40	Executive	10.0.29.128	255.255.255.192 (/26)	62

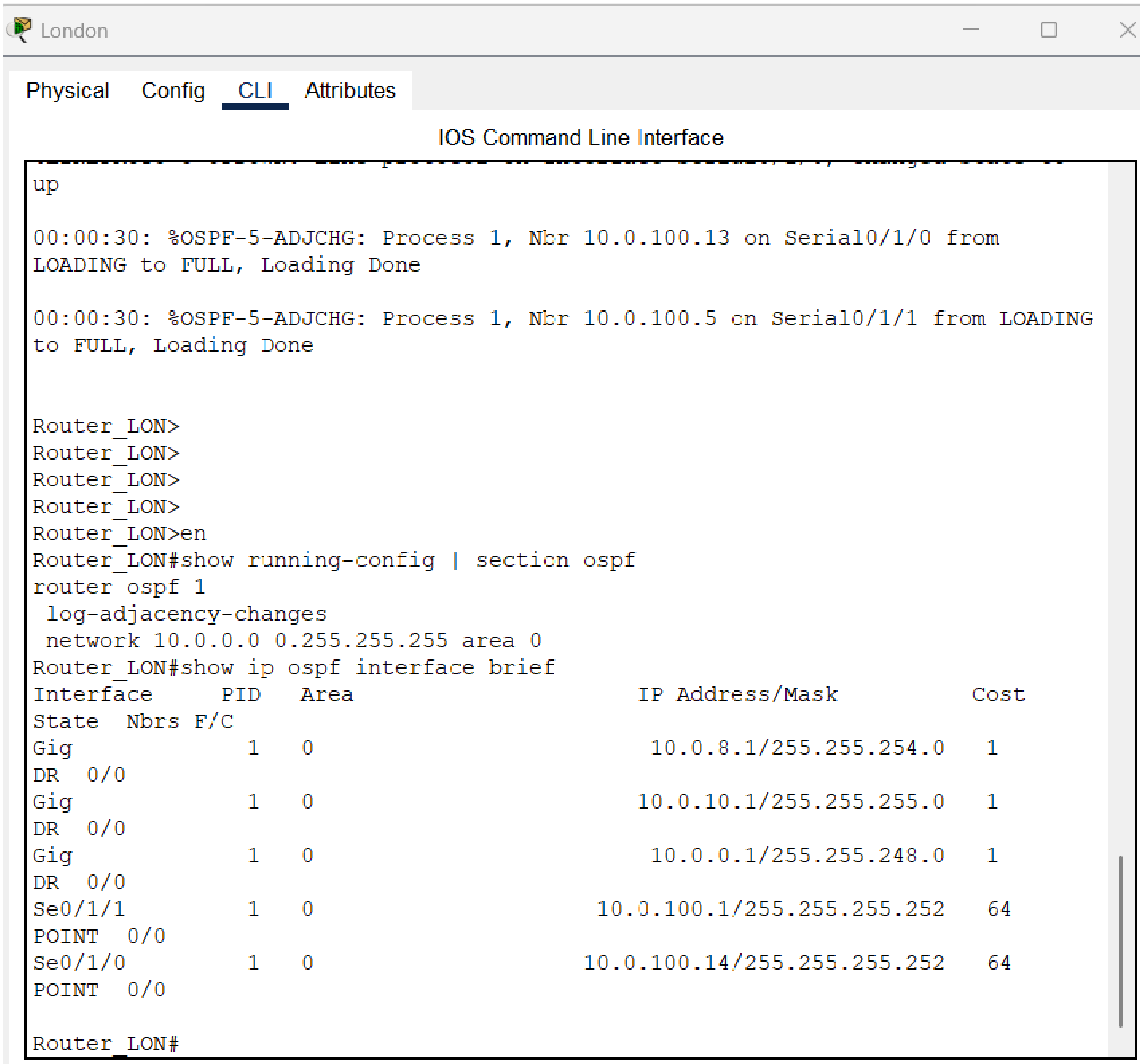
Leeds				
VLAN	**Purpose**	**Network Address**	**Subnet Mask**	**Hosts**
VLAN 50	Employees	10.0.24.0	255.255.254.0 (/23)	510
VLAN 30	Finance	10.0.26.0	255.255.255.0 (/24)	254
VLAN 40	Executive	10.0.27.0	255.255.255.128 (/25)	126

---

3. Routing Strategy

3.1. Routing Protocol Selection

- OSPF is implemented as the dynamic routing protocol across all routers to ensure efficient and scalable route exchange between sites. Each router participates in a single OSPF area, advertising its connected networks and learning routes dynamically. This eliminates the need for static routing and allows automatic adaptation to network changes. OSPF ensures optimal path selection and enables full connectivity between all VLANs across different sites.



London

Physical Config CLI Attributes

IOS Command Line Interface

```
up

00:00:30: %OSPF-5-ADJCHG: Process 1, Nbr 10.0.100.13 on Serial0/1/0 from
LOADING to FULL, Loading Done

00:00:30: %OSPF-5-ADJCHG: Process 1, Nbr 10.0.100.5 on Serial0/1/1 from LOADING
to FULL, Loading Done


Router_LON>
Router_LON>
Router_LON>
Router_LON>
Router_LON>en
Router_LON#show running-config | section ospf
router ospf 1
 log-adjacency-changes
 network 10.0.0.0 0.255.255.255 area 0
Router_LON#show ip ospf interface brief
Interface      PID   Area                     IP Address/Mask          Cost
State  Nbrs F/C
Gig              1    0                        10.0.8.1/255.255.254.0    1
DR   0/0
Gig              1    0                       10.0.10.1/255.255.255.0    1
DR   0/0
Gig              1    0                        10.0.0.1/255.255.248.0    1
DR   0/0
Se0/1/1          1    0                    10.0.100.1/255.255.255.252    64
POINT  0/0
Se0/1/0          1    0                   10.0.100.14/255.255.255.252    64
POINT  0/0

Router_LON#
```

<<London OSPF Configuration>>

```
Router_LON#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 200.0.0.2 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 23 subnets, 8 masks
C       10.0.0.0/21 is directly connected, GigabitEthernet0/0/0.50
L       10.0.0.1/32 is directly connected, GigabitEthernet0/0/0.50
C       10.0.8.0/23 is directly connected, GigabitEthernet0/0/0.30
L       10.0.8.1/32 is directly connected, GigabitEthernet0/0/0.30
C       10.0.10.0/24 is directly connected, GigabitEthernet0/0/0.40
L       10.0.10.1/32 is directly connected, GigabitEthernet0/0/0.40
O       10.0.16.0/22 [110/65] via 10.0.100.13, 01:07:08, Serial0/1/0
O       10.0.20.0/23 [110/65] via 10.0.100.13, 01:07:08, Serial0/1/0
O       10.0.22.0/24 [110/65] via 10.0.100.13, 01:07:08, Serial0/1/0
O       10.0.24.0/23 [110/65] via 10.0.100.2, 01:07:08, Serial0/1/1
O       10.0.26.0/24 [110/65] via 10.0.100.2, 01:07:08, Serial0/1/1
 --More--
```

<<London Routing Table>>

```
O       10.0.26.0/24 [110/65] via 10.0.100.2, 01:07:08, Serial0/1/1
O       10.0.27.0/25 [110/65] via 10.0.100.2, 01:07:08, Serial0/1/1
O       10.0.28.0/24 [110/129] via 10.0.100.13, 01:07:08, Serial0/1/0
                     [110/129] via 10.0.100.2, 01:07:08, Serial0/1/1
O       10.0.29.0/25 [110/129] via 10.0.100.13, 01:07:08, Serial0/1/0
                     [110/129] via 10.0.100.2, 01:07:08, Serial0/1/1
O       10.0.29.128/26 [110/129] via 10.0.100.13, 01:07:08, Serial0/1/0
                       [110/129] via 10.0.100.2, 01:07:08, Serial0/1/1
C       10.0.50.0/24 is directly connected, GigabitEthernet0/0/0.60
L       10.0.50.1/32 is directly connected, GigabitEthernet0/0/0.60
C       10.0.100.0/30 is directly connected, Serial0/1/1
L       10.0.100.1/32 is directly connected, Serial0/1/1
O       10.0.100.4/30 [110/128] via 10.0.100.2, 01:07:08, Serial0/1/1
O       10.0.100.8/30 [110/128] via 10.0.100.13, 01:07:08, Serial0/1/0
C       10.0.100.12/30 is directly connected, Serial0/1/0
L       10.0.100.14/32 is directly connected, Serial0/1/0
     200.0.0.0/24 is variably subnetted, 2 subnets, 2 masks
C       200.0.0.0/30 is directly connected, GigabitEthernet0/0/1
L       200.0.0.1/32 is directly connected, GigabitEthernet0/0/1
S*   0.0.0.0/0 [1/0] via 200.0.0.2
```

<<London Routing Table>>

```
Router_LEE#show running-config | section ospf
router ospf 1
 log-adjacency-changes
 network 10.0.0.0 0.255.255.255 area 0
Router_LEE#show ip os
Router_LEE#show ip ospf in
Router_LEE#show ip ospf interface brief
Interface      PID   Area                    IP Address/Mask          Cost
State  Nbrs F/C
Gig             1    0                       10.0.26.1/255.255.255.0   1
DR   0/0
Gig             1    0                     10.0.27.1/255.255.255.128   1
DR   0/0
Gig             1    0                       10.0.24.1/255.255.254.0   1
DR   0/0
Se0/1/1         1    0                    10.0.100.5/255.255.255.252   64
POINT  0/0
Se0/1/0         1    0                    10.0.100.2/255.255.255.252   64
POINT  0/0
```

<<Leeds OSPF Configuration>>

```
Router_LEE#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 22 subnets, 8 masks
O       10.0.0.0/21 [110/65] via 10.0.100.1, 00:07:21, Serial0/1/0
O       10.0.8.0/23 [110/65] via 10.0.100.1, 00:07:21, Serial0/1/0
O       10.0.10.0/24 [110/65] via 10.0.100.1, 00:07:21, Serial0/1/0
O       10.0.16.0/22 [110/129] via 10.0.100.6, 00:07:21, Serial0/1/1
                     [110/129] via 10.0.100.1, 00:07:21, Serial0/1/0
O       10.0.20.0/23 [110/129] via 10.0.100.6, 00:07:21, Serial0/1/1
                     [110/129] via 10.0.100.1, 00:07:21, Serial0/1/0
O       10.0.22.0/24 [110/129] via 10.0.100.6, 00:07:21, Serial0/1/1
                     [110/129] via 10.0.100.1, 00:07:21, Serial0/1/0
C       10.0.24.0/23 is directly connected, GigabitEthernet0/0/0.50
L       10.0.24.1/32 is directly connected, GigabitEthernet0/0/0.50
C       10.0.26.0/24 is directly connected, GigabitEthernet0/0/0.30
L       10.0.26.1/32 is directly connected, GigabitEthernet0/0/0.30
C       10.0.27.0/25 is directly connected, GigabitEthernet0/0/0.40
L       10.0.27.1/32 is directly connected, GigabitEthernet0/0/0.40
O       10.0.28.0/24 [110/65] via 10.0.100.6, 00:07:31, Serial0/1/1
O       10.0.29.0/25 [110/65] via 10.0.100.6, 00:07:31, Serial0/1/1
O       10.0.29.128/26 [110/65] via 10.0.100.6, 00:07:31, Serial0/1/1
O       10.0.50.0/24 [110/65] via 10.0.100.1, 00:07:21, Serial0/1/0
C       10.0.100.0/30 is directly connected, Serial0/1/0
L       10.0.100.2/32 is directly connected, Serial0/1/0
C       10.0.100.4/30 is directly connected, Serial0/1/1
L       10.0.100.5/32 is directly connected, Serial0/1/1
O       10.0.100.8/30 [110/128] via 10.0.100.6, 00:07:31, Serial0/1/1
O       10.0.100.12/30 [110/128] via 10.0.100.1, 00:07:21, Serial0/1/0

Router_LEE#
```

<<Leeds Routing Table>>

```
Router_LIV#show running-config | section access-list
access-list 150 permit ip 10.0.28.0 0.0.0.255 10.0.16.0 0.0.3.255
access-list 150 permit ip 10.0.28.0 0.0.0.255 10.0.24.0 0.0.1.255
access-list 150 permit ip 10.0.28.0 0.0.0.255 10.0.0.0 0.0.7.255
access-list 150 permit udp any eq bootps any eq bootpc
access-list 150 permit udp any eq bootpc any eq bootps
access-list 150 deny ip 10.0.28.0 0.0.0.255 any
Router_LIV#show ip os
Router_LIV#show ip ospf i
Router_LIV#show ip ospf interface brief
Interface      PID   Area             IP Address/Mask         Cost  State  Nbrs F/C
Gig              1   0              10.0.29.1/255.255.255.128   1      DR  0/0
Gig              1   0            10.0.29.129/255.255.255.192   1      DR  0/0
Gig              1   0                10.0.28.1/255.255.255.0   1      DR  0/0
Se0/1/0          1   0             10.0.100.9/255.255.255.252   64  POINT  0/0
Se0/1/1          1   0             10.0.100.6/255.255.255.252   64  POINT  0/0

Router_LIV#
```

<<Liverpool OSPF Configuration>>

```
Router_LIV#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 22 subnets, 8 masks
O       10.0.0.0/21 [110/129] via 10.0.100.10, 00:13:40, Serial0/1/0
                    [110/129] via 10.0.100.5, 00:13:40, Serial0/1/1
O       10.0.8.0/23 [110/129] via 10.0.100.10, 00:13:40, Serial0/1/0
                    [110/129] via 10.0.100.5, 00:13:40, Serial0/1/1
O       10.0.10.0/24 [110/129] via 10.0.100.10, 00:13:40, Serial0/1/0
                     [110/129] via 10.0.100.5, 00:13:40, Serial0/1/1
O       10.0.16.0/22 [110/65] via 10.0.100.10, 00:13:50, Serial0/1/0
O       10.0.20.0/23 [110/65] via 10.0.100.10, 00:13:50, Serial0/1/0
O       10.0.22.0/24 [110/65] via 10.0.100.10, 00:13:50, Serial0/1/0
O       10.0.24.0/23 [110/65] via 10.0.100.5, 00:13:50, Serial0/1/1
O       10.0.26.0/24 [110/65] via 10.0.100.5, 00:13:50, Serial0/1/1
O       10.0.27.0/25 [110/65] via 10.0.100.5, 00:13:50, Serial0/1/1
C       10.0.28.0/24 is directly connected, GigabitEthernet0/0/0.50
L       10.0.28.1/32 is directly connected, GigabitEthernet0/0/0.50
C       10.0.29.0/25 is directly connected, GigabitEthernet0/0/0.30
L       10.0.29.1/32 is directly connected, GigabitEthernet0/0/0.30
C       10.0.29.128/26 is directly connected, GigabitEthernet0/0/0.40
L       10.0.29.129/32 is directly connected, GigabitEthernet0/0/0.40
O       10.0.50.0/24 [110/129] via 10.0.100.10, 00:13:40, Serial0/1/0
                     [110/129] via 10.0.100.5, 00:13:40, Serial0/1/1
O       10.0.100.0/30 [110/128] via 10.0.100.5, 00:13:50, Serial0/1/1
C       10.0.100.4/30 is directly connected, Serial0/1/1
L       10.0.100.6/32 is directly connected, Serial0/1/1
C       10.0.100.8/30 is directly connected, Serial0/1/0
L       10.0.100.9/32 is directly connected, Serial0/1/0
O       10.0.100.12/30 [110/128] via 10.0.100.10, 00:13:50, Serial0/1/0
```

<<Liverpool Routing Table>>

```
Router-LEI#show running-config | section access-list
access-list 150 permit ip 10.0.16.0 0.0.3.255 10.0.28.0 0.0.0.255
access-list 150 permit ip 10.0.16.0 0.0.3.255 10.0.24.0 0.0.1.255
access-list 150 permit ip 10.0.16.0 0.0.3.255 10.0.0.0 0.0.7.255
access-list 150 permit udp any eq bootpc any eq bootps
access-list 150 permit udp any eq bootps any eq bootpc
access-list 150 deny ip 10.0.16.0 0.0.3.255 any
Router-LEI#show ip os
Router-LEI#show ip ospf i
Router-LEI#show ip ospf interface brief
Interface      PID   Area              IP Address/Mask         Cost  State  Nbrs F/C
Gig              1   0                 10.0.20.1/255.255.254.0   1      DR  0/0
Gig              1   0                 10.0.22.1/255.255.255.0   1      DR  0/0
Gig              1   0                 10.0.16.1/255.255.252.0   1      DR  0/0
Se0/1/1          1   0              10.0.100.10/255.255.255.252  64  POINT  0/0
Se0/1/0          1   0              10.0.100.13/255.255.255.252  64  POINT  0/0

Router-LEI#
```

<<Leicester OSPF Configuration>>

```
Router-LEI#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 22 subnets, 8 masks
O       10.0.0.0/21 [110/65] via 10.0.100.14, 00:19:04, Serial0/1/0
O       10.0.8.0/23 [110/65] via 10.0.100.14, 00:19:04, Serial0/1/0
O       10.0.10.0/24 [110/65] via 10.0.100.14, 00:19:04, Serial0/1/0
C       10.0.16.0/22 is directly connected, GigabitEthernet0/0/0.50
L       10.0.16.1/32 is directly connected, GigabitEthernet0/0/0.50
C       10.0.20.0/23 is directly connected, GigabitEthernet0/0/0.30
L       10.0.20.1/32 is directly connected, GigabitEthernet0/0/0.30
C       10.0.22.0/24 is directly connected, GigabitEthernet0/0/0.40
L       10.0.22.1/32 is directly connected, GigabitEthernet0/0/0.40
O       10.0.24.0/23 [110/129] via 10.0.100.9, 00:19:04, Serial0/1/1
                     [110/129] via 10.0.100.14, 00:19:04, Serial0/1/0
O       10.0.26.0/24 [110/129] via 10.0.100.9, 00:19:04, Serial0/1/1
                     [110/129] via 10.0.100.14, 00:19:04, Serial0/1/0
O       10.0.27.0/25 [110/129] via 10.0.100.9, 00:19:04, Serial0/1/1
                     [110/129] via 10.0.100.14, 00:19:04, Serial0/1/0
O       10.0.28.0/24 [110/65] via 10.0.100.9, 00:19:04, Serial0/1/1
O       10.0.29.0/25 [110/65] via 10.0.100.9, 00:19:04, Serial0/1/1
O       10.0.29.128/26 [110/65] via 10.0.100.9, 00:19:04, Serial0/1/1
O       10.0.50.0/24 [110/65] via 10.0.100.14, 00:19:04, Serial0/1/0
O       10.0.100.0/30 [110/128] via 10.0.100.14, 00:19:04, Serial0/1/0
O       10.0.100.4/30 [110/128] via 10.0.100.9, 00:19:04, Serial0/1/1
C       10.0.100.8/30 is directly connected, Serial0/1/1
L       10.0.100.10/32 is directly connected, Serial0/1/1
C       10.0.100.12/30 is directly connected, Serial0/1/0
L       10.0.100.13/32 is directly connected, Serial0/1/0
```

<<Leicester Routing Table>>

3.2. IP Addressing Scheme

A Variable Length Subnet Masking (VLSM) scheme is implemented to efficiently allocate IP addresses based on departmental requirements. Each VLAN is assigned a dedicated subnet, with larger subnets allocated to departments with more hosts, such as Employees, and smaller subnets to Finance and Executive departments. This approach minimizes IP wastage and improves routing efficiency. The addressing scheme follows a structured private IP range (10.0.0.0/8), ensuring scalability and consistency across all sites while supporting inter-site communication via OSPF.

Device	Interface	IP Address	Subnet Mask
Router_LEE	GigabitEthernet 0/1	N/A	N/A
	Serial 0/1/0	10.0.100.2	255.255.255.252
	Serial 0/1/1	100.0.100.5	255.255.255.252
Router_LEI	GigabitEthernet 0/1	N/A	N/A
	Serial 0/1/ 0	10.0.100.13	255.255.255.252
	Serial 0/1/1	10.0.100.10	255.255.255.252
Router_LIV	GigabitEthernet 0/1	N/A	N/A
	Serial 0/1/0	10.0.100.9	255.255.255.252
	Serial 0/1/1	10.0.100.6	255.255.255.252

London

Device	Interface	IP Address	Subnet Mask
Router_LON	G0/0/0	N/C	N/C
	G0/0/1	N/C	N/C
	G0/0/0.30	10.0.8.1	255.255.254.0
	G0/0/0.40	10.0.10.1	255.255.255.0
	G0/0/0.50	10.0.0.1	255.255.248.0
	G0/0/0.60	10.0.50.1	255.255.255.0
	Se0/1/0	10.0.100.14	255.255.255.252
	Se0/1/1	10.0.100.1	255.255.255.252

LON-SW1	Vlan 30	10.0.8.1	255.255.254.0
	Vlan 40	10.0.10.1	255.255.255.0
	Vlan 50	10.0.0.1	255.255.248.0
	VLAN 60	10.0.50.1	255.255.255.0
PC-5	F0/4	DHCP	DHCP
PC-4	F0/3	DHCP	DHCP
Laptop-2	F0/2	DHCP	DHCP
Server 0	F0/5	10.0.50.2	255.255.255.0

Leeds

Device	Interface	IP Address	Subnet Mask
Router_LON	G0/0/0	N/C	N/C
	G0/0/1	N/C	N/C
	G0/0/0.30	10.0.26.1	255.255.255.0
	G0/0/0.40	10.0.27.1	255.255.255.128
	G0/0/0.50	10.0.24.1	255.255.254.0
	Se0/1/0	10.0.100.2	255.255.255.252
	Se0/1/1	10.0.100.5	255.255.255.252
LON-SW1	Vlan 30	10.0.26.1	255.255.255.0
	Vlan 40	10.0.27.1	255.255.255.128
	Vlan 50	10.0.24.1	255.255.254.0
PC-3	F0/4	DHCP	DHCP
PC-2	F0/3	DHCP	DHCP
Laptop-1	F0/2	DHCP	DHCP

Liverpool

Device	Interface	IP Address	Subnet Mask
Router_LON	G0/0/0	N/C	N/C
	G0/0/1	N/C	N/C
	G0/0/0.30	10.0.29.1	255.255.255.128
	G0/0/0.40	10.0.29.129	255.255.255.192
	G0/0/0.50	10.0.28.1	255.255.255.0
	Se0/1/0	10.0.100.9	255.255.255.252

	Se0/1/1	10.0.100.6	255.255.255.252
LON-SW1	Vlan 30	10.0.29.1	255.255.254.0
	Vlan 40	10.0.29.129	255.255.255.0
	Vlan 50	10.0.28.1	255.255.248.0
PC-9	F0/4	DHCP	DHCP
PC-8	F0/3	DHCP	DHCP
Laptop-4	F0/2	DHCP	DHCP

Leicester

Device	Interface	IP Address	Subnet Mask
Router_LON	G0/0/0	N/C	N/C
	G0/0/1	N/C	N/C
	G0/0/0.30	10.0.20.1	255.255.255.128
	G0/0/0.40	10.0.22.1	255.255.255.0
	G0/0/0.50	10.0.16.1	255.255.252.0
	Se0/1/0	10.0.100.13	255.255.255.252
	Se0/1/1	10.0.100.10	255.255.255.252
LON-SW1	Vlan 30	10.0.20.1	255.255.255.128
	Vlan 40	10.0.22.1	255.255.255.0
	Vlan 50	10.0.16.1	255.255.252.0
PC-7	F0/4	DHCP	DHCP
PC-6	F0/3	DHCP	DHCP
Laptop-3	F0/2	DHCP	DHCP

---

4. Switching & DHCP Implementation

4.1. VLAN Configuration

- VLAN assignment using Layer 2 switches.
- Inter-VLAN routing via layer 3 switches.

LON-SW1

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state
to up

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state
to up


LON-SW1>
LON-SW1>
LON-SW1>en
LON-SW1#show vlan br
LON-SW1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
30   Finance                          active    Fa0/2
40   Executive                        active    Fa0/3
50   Employees                        active    Fa0/4
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

<<London Switch VLAN Configuration>>

```
LON-SW1#show running-config | section interface
interface FastEthernet0/1
 switchport mode trunk
interface FastEthernet0/2
 switchport access vlan 30
 switchport mode access
interface FastEthernet0/3
 switchport access vlan 40
 switchport mode access
interface FastEthernet0/4
 switchport access vlan 50
 switchport mode access
interface FastEthernet0/5
interface FastEthernet0/6
interface FastEthernet0/7
interface FastEthernet0/8
interface FastEthernet0/9
interface FastEthernet0/10
interface FastEthernet0/11
interface FastEthernet0/12
interface FastEthernet0/13
interface FastEthernet0/14
interface FastEthernet0/15
interface FastEthernet0/16
interface FastEthernet0/17
interface FastEthernet0/18
interface FastEthernet0/19
interface FastEthernet0/20
interface FastEthernet0/21
interface FastEthernet0/22
interface FastEthernet0/23
interface FastEthernet0/24
interface GigabitEthernet0/1
interface GigabitEthernet0/2
interface Vlan1
 no ip address
 shutdown
LON-SW1#
```

<<London Switch Access Ports>>

```
LON-SW1#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       1-1005

Port        Vlans allowed and active in management domain
Fa0/1       1,30,40,50

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       1,30,40,50

LON-SW1#
```

- **<<London Trunk Link Configuration between Switch and Router>>**

```
LEE-SW1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
30   Finance                          active    Fa0/2
40   Executive                        active    Fa0/3
50   Employees                        active    Fa0/4
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
LEE-SW1#
```

<<Leeds VLAN Configuration>>

```
LIV-SW1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
30   Finance                          active    Fa0/2
40   Executive                        active    Fa0/3
50   Employees                        active    Fa0/4
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
LIV-SW1#
```

<<Liverpool VLAN Configuration>>

```
LEI-SW1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
30   Finance                          active    Fa0/2
40   Executive                        active    Fa0/3
50   Employees                        active    Fa0/4
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
LEI-SW1#
```

<<Leicester VLAN Configuration>>

4.2. DHCP Implementation

Each site is configured with DHCP services on the router to automatically assign IP addresses to devices within each VLAN. Separate DHCP pools are created for Finance, Executive, and Employees networks, specifying network ranges, default gateways, and excluded addresses. This reduces manual configuration errors and simplifies network management. Proper ACL configuration ensures that DHCP traffic (UDP ports 67 and 68) is permitted, preventing address assignment failures.

```
Router_LON>show ip dhcp pool

Pool LON_EMP :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 2046
 Leased addresses               : 1
 Excluded addresses             : 3
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 10.0.0.1             10.0.0.1         - 10.0.7.254        1    / 3      / 2046

Pool LON_EXEC :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 1
 Excluded addresses             : 3
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 10.0.10.1            10.0.10.1        - 10.0.10.254       1    / 3      / 254

Pool LON_FIN :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 510
 Leased addresses               : 1
 Excluded addresses             : 3
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 10.0.8.1             10.0.8.1         - 10.0.9.254        1    / 3      / 510
Router_LON>
```

<<London DHCP Pool>>

- ```
  Router_LON#show running-config | section dhcp
  ip dhcp excluded-address 10.0.8.1
  ip dhcp excluded-address 10.0.10.1
  ip dhcp excluded-address 10.0.0.1
  ip dhcp pool LON_EMP
   network 10.0.0.0 255.255.248.0
   default-router 10.0.0.1
   dns-server 8.8.8.8
  ip dhcp pool LON_EXEC
   network 10.0.10.0 255.255.255.0
   default-router 10.0.10.1
   dns-server 8.8.8.8
  ip dhcp pool LON_FIN
   network 10.0.8.0 255.255.254.0
   default-router 10.0.8.1
   dns-server 8.8.8.8
  Router_LON#
  ```

<<London DHCP Configuration>>

- ```
  Router_LEE#show ip dhcp pool

  Pool LEEDS_EMP :
   Utilization mark (high/low)    : 100 / 0
   Subnet size (first/next)       : 0 / 0
   Total addresses                : 510
   Leased addresses               : 1
   Excluded addresses             : 3
   Pending event                  : none

   1 subnet is currently in the pool
   Current index        IP address range                    Leased/Excluded/Total
   10.0.24.1            10.0.24.1        - 10.0.25.254       1    / 3     / 510

  Pool LEEDS_FIN :
   Utilization mark (high/low)    : 100 / 0
   Subnet size (first/next)       : 0 / 0
   Total addresses                : 254
   Leased addresses               : 1
   Excluded addresses             : 3
   Pending event                  : none

   1 subnet is currently in the pool
   Current index        IP address range                    Leased/Excluded/Total
   10.0.26.1            10.0.26.1        - 10.0.26.254       1    / 3     / 254

  Pool LEEDS_EXEC :
   Utilization mark (high/low)    : 100 / 0
   Subnet size (first/next)       : 0 / 0
   Total addresses                : 126
   Leased addresses               : 1
   Excluded addresses             : 3
   Pending event                  : none

   1 subnet is currently in the pool
   Current index        IP address range                    Leased/Excluded/Total
   10.0.27.1            10.0.27.1        - 10.0.27.126       1    / 3     / 126
  Router_LEE#
  ```

<<Leeds DHCP Pool>>

```
Router_LIV#show ip dhcp pool

Pool LIV_FIN :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 126
 Leased addresses               : 1
 Excluded addresses             : 3
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 10.0.29.1            10.0.29.1        - 10.0.29.126       1    / 3      / 126

Pool LIV_EXEC :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 62
 Leased addresses               : 1
 Excluded addresses             : 3
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 10.0.29.129          10.0.29.129      - 10.0.29.190       1    / 3      / 62

Pool LIV_EMP :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 1
 Excluded addresses             : 3
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 10.0.28.1            10.0.28.1        - 10.0.28.254       1    / 3      / 254
Router_LIV#
```

<<Liverpool DHCP Pool>>

```
Router-LEI#show ip dhcp pool

Pool LEIC_FIN :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 510
 Leased addresses               : 1
 Excluded addresses             : 3
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 10.0.20.1            10.0.20.1        - 10.0.21.254       1     / 3      / 510

Pool LEIC_EXEC :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 1
 Excluded addresses             : 3
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 10.0.22.1            10.0.22.1        - 10.0.22.254       1     / 3      / 254

Pool LEIC_EMP :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 1022
 Leased addresses               : 1
 Excluded addresses             : 3
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 10.0.16.1            10.0.16.1        - 10.0.19.254       1     / 3      / 1022
Router-LEI#
```

<<Leicester DHCP Pool>>

```
Router_LON#show ip dhcp binding
IP address          Client-ID/              Lease expiration        Type
                    Hardware address
10.0.0.2            0005.5EA4.C120           --                     Automatic
10.0.10.2           0002.1675.14AA           --                     Automatic
10.0.8.2            0060.709E.421B           --                     Automatic
Router_LON#
```

<<London DHCP binding>>

```
Router_LEE#show ip dhcp binding
IP address          Client-ID/              Lease expiration        Type
                    Hardware address
10.0.24.2           0001.C7C6.A6A8           --                     Automatic
10.0.26.2           0040.0B42.2D47           --                     Automatic
10.0.27.2           000B.BE02.D4EB           --                     Automatic
Router_LEE#
```

<<Leeds DHCP binding>>

```
Router-LIV#show ip dhcp binding
IP address        Client-ID/              Lease expiration        Type
                  Hardware address
10.0.29.2         0001.42CC.E038           --                     Automatic
10.0.29.130       0030.A317.111B           --                     Automatic
10.0.28.2         0002.4ACA.2490           --                     Automatic
Router-LIV#
```

<<Liverpool DHCP binding>>

```
Router-LEI#show ip dhcp binding
IP address        Client-ID/              Lease expiration        Type
                  Hardware address
10.0.20.2         0060.3E47.6ACB           --                     Automatic
10.0.22.2         0001.96C5.524A           --                     Automatic
10.0.16.2         00D0.D3C9.1A75           --                     Automatic
Router-LEI#
```

<<Leicester DHCP binding>>

5. Network Security Implementation

5.1. Access Control & Security Policies

Extended ACLs are implemented to enforce security policies across the network. Employees are restricted to communicating only with employees in other sites, while Executive users are granted broader access to all departments. Internet access is limited to the London site, with other sites restricted through network design and ACL policies. ACLs are carefully ordered to allow essential services such as DHCP before applying deny rules, ensuring proper functionality while maintaining security.

Network Address Translation (NAT) is configured exclusively on the London router to enable internal devices to access external networks using a public IP address. The NAT overload (PAT) technique is used to translate multiple private IP addresses into a single public IP. A default route is configured to forward external traffic to the ISP router. This design ensures that only the London site has internet access, while other sites are restricted due to the absence of NAT configuration.

```
Router_LEE#show running-config | section access-list
access-list 150 permit ip 10.0.24.0 0.0.1.255 10.0.0.0 0.0.7.255
access-list 150 permit ip 10.0.24.0 0.0.1.255 10.0.28.0 0.0.0.255
access-list 150 permit ip 10.0.24.0 0.0.1.255 10.0.16.0 0.0.3.255
access-list 150 permit udp any eq bootps any eq bootpc
access-list 150 permit udp any eq bootpc any eq bootps
access-list 150 deny ip 10.0.24.0 0.0.1.255 any
Router LEE#
```

<<Leeds ACLs>>

- ```
  Router_LIV#show running-config | section access-list
  access-list 150 permit ip 10.0.28.0 0.0.0.255 10.0.16.0 0.0.3.255
  access-list 150 permit ip 10.0.28.0 0.0.0.255 10.0.24.0 0.0.1.255
  access-list 150 permit ip 10.0.28.0 0.0.0.255 10.0.0.0 0.0.7.255
  access-list 150 permit udp any eq bootps any eq bootpc
  access-list 150 permit udp any eq bootpc any eq bootps
  access-list 150 deny ip 10.0.28.0 0.0.0.255 any
  Router_LIV#
  ```

<<Liverpool ACLs>>

- ```
  Router_LON#show running-config | section access-list
  access-list 160 permit ip any 10.0.50.0 0.0.0.255
  access-list 160 deny ip any 10.0.0.0 0.255.255.255
  access-list 1 permit 10.0.0.0 0.0.31.255
  access-list 150 permit ip 10.0.0.0 0.0.7.255 10.0.24.0 0.0.1.255
  access-list 150 permit ip 10.0.0.0 0.0.7.255 10.0.28.0 0.0.0.255
  access-list 150 permit ip 10.0.0.0 0.0.7.255 10.0.16.0 0.0.3.255
  access-list 150 permit udp any eq bootps any eq bootpc
  access-list 150 permit udp any eq bootpc any eq bootps
  access-list 150 deny ip 10.0.0.0 0.0.7.255 any
  ```

<<London ACLs>>

- ```
  Router-LEI#show running-config | section access-list
  access-list 150 permit ip 10.0.16.0 0.0.3.255 10.0.28.0 0.0.0.255
  access-list 150 permit ip 10.0.16.0 0.0.3.255 10.0.24.0 0.0.1.255
  access-list 150 permit ip 10.0.16.0 0.0.3.255 10.0.0.0 0.0.7.255
  access-list 150 permit udp any eq bootpc any eq bootps
  access-list 150 permit udp any eq bootps any eq bootpc
  access-list 150 deny ip 10.0.16.0 0.0.3.255 any
  ```

<<Leicester ACLs>>

5.2. Device Hardening Measures

Encrypted Basic device security is implemented by configuring SSH for secure remote access, disabling insecure protocols such as Telnet, and applying login authentication using local user accounts. A message-of-the-day (MOTD) banner is configured to warn unauthorized users, and passwords are encrypted to protect sensitive information. These measures enhance the overall security posture of the network and align with best practices for enterprise environments.
  ```

```
Router_LON#show running-config | include service password-encryption
service password-encryption
Router_LON#show ru
Router_LON#show running-config | section line
line con 0
 exec-timeout 5 0
 password 7 0873181C5A4B5345
 login
line aux 0
line vty 0 4
 exec-timeout 5 0
 password 7 0822455D0A16544541
 login
 transport input ssh
Router LON#
```

<<London Router Security Measures>>

```
Router-LEI#show running-config | include service password-encryption
service password-encryption
Router-LEI#show ru
Router-LEI#show running-config  | section line
line con 0
 exec-timeout 5 0
 password 7 0873181C5A4B5345
 login
line aux 0
line vty 0 4
 exec-timeout 5 0
 password 7 0873181C5A4B5345
 login
 transport input ssh
Router-LEI#
```

<<Leicester Router Security Measures>>

```
Router_LIV#show running-config | section service password-encryption
service password-encryption
Router_LIV#show running-config | section line
line con 0
 password 7 0873181C5A4B5345
 login
line aux 0
line vty 0 4
 password 7 0873181C5A4B5345
 login
 transport input ssh
Router_LIV#
```

<<Liverpool Router Security Measures>>

```
Router_LEE#show running-config | section line
line con 0
 exec-timeout 5 0
 password 7 0873181C5A4B5345
 login
line aux 0
line vty 0 4
 exec-timeout 5 0
 password 7 0873181C5A4B5345
 login
 transport input ssh
Router_LEE#show running-config | section service password-encryption
service password-encryption
Router_LEE#
```

<<Leeds Router Security Measures>>

---

6. Testing & Validation

6.1. Connectiviy Testing

- Ping and Traceroute commands to verify network reachability.
- OSPF neighbor verification through CLI commands.
- The VLAN 50 (Employees) is restricted to finance and executive, i.e, it has only access to other VLAN 50s.

PDU List Window

Fire	Last Status	Source	Destination	Type	Color	Time(se	Periodic	Num	Edit	Delete
	Successful	Lon...	Leicester	ICMP		0.000	N	0	(edit)	
	Successful	Lon...	Leeds	ICMP		0.000	N	1	(edit)	
	Successful	Lon...	Liverpool	ICMP		0.000	N	2	(edit)	
	Successful	Leic...	Liverpool	ICMP		0.000	N	3	(edit)	
	Successful	Leic...	Leeds	ICMP		0.000	N	4	(edit)	
	Successful	Live...	Leeds	ICMP		0.000	N	5	(edit)	

<<Ping test between Routers>>

PDU List Window

Fire	Last Status	Source	Destination	Type	Color	Time(se	Periodic	Num	Edit	Delete
	Successful	Lapt...	PC-4	ICMP		0.000	N	0	(edit)	
	Failed	Lapt...	PC-5	ICMP		0.000	N	1	(edit)	
	Successful	Lapt...	DMZ	ICMP		0.000	N	2	(edit)	
	Successful	Lapt...	London	ICMP		0.000	N	3	(edit)	
	Failed	PC-4	PC-5	ICMP		0.000	N	4	(edit)	
	Successful	PC-4	DMZ	ICMP		0.000	N	5	(edit)	
	Successful	PC-4	London	ICMP		0.000	N	6	(edit)	
	Failed	PC-5	DMZ	ICMP		0.000	N	7	(edit)	
	Failed	PC-5	London	ICMP		0.000	N	8	(edit)	
	Successful	DMZ	London	ICMP		0.000	N	9	(edit)	

<<Ping test in London LAN>>

Fire	Last Status	Source	Destination	Type	Color	Time(se	Periodic	Num	Edit	Delete
	Failed	Lapt...	PC-2	ICMP		0.000	N	0	(edit)	
	Failed	Lapt...	PC-3	ICMP		0.000	N	1	(edit)	
	Successful	Lapt...	Leeds	ICMP		0.000	N	2	(edit)	
	Failed	PC-2	PC-3	ICMP		0.000	N	3	(edit)	
	Successful	PC-2	Leeds	ICMP		0.000	N	4	(edit)	
	Failed	PC-3	Leeds	ICMP		0.000	N	5	(edit)	

<<Ping test in Leeds LAN>>

Fire	Last Status	Source	Destination	Type	Color	Time(se	Periodic	Num	Edit	Delete
	Failed	PC-9	PC-8	ICMP		0.000	N	0	(edit)	
	Failed	PC-9	Laptop-4	ICMP		0.000	N	1	(edit)	
	Failed	PC-9	Liverpool	ICMP		0.000	N	2	(edit)	
	Failed	PC-8	Laptop-4	ICMP		0.000	N	3	(edit)	
	Successful	PC-8	Liverpool	ICMP		0.000	N	4	(edit)	
	Successful	Lapt...	Liverpool	ICMP		0.000	N	5	(edit)	

<<Ping test in Liverpool LAN>>

PDU List Window

Fire	Last Status	Source	Destination	Type	Color	Time(se	Periodic	Num	Edit	Delete
	Successful	Lapt...	Leicester	ICMP		0.000	N	0	(edit)	
	Successful	Lapt...	PC-6	ICMP		0.000	N	1	(edit)	
	Failed	Lapt...	PC-7	ICMP		0.000	N	2	(edit)	
	Failed	PC-6	PC-7	ICMP		0.000	N	3	(edit)	
	Successful	PC-6	Leicester	ICMP		0.000	N	4	(edit)	
	Failed	PC-7	Leicester	ICMP		0.000	N	5	(edit)	

<<Ping test in Leicester LAN>>

PDU List Window

Fire	Last Status	Source	Destination	Type	Color	Time(se	Periodic	Num	Edit	Delete
	Successful	PC-5	PC-7	ICMP		0.000	N	0	(edit)	
	Successful	PC-5	PC-3	ICMP		0.000	N	1	(edit)	
	Successful	PC-5	PC-9	ICMP		0.000	N	2	(edit)	
	Successful	PC-7	PC-3	ICMP		0.000	N	3	(edit)	
	Successful	PC-7	PC-9	ICMP		0.000	N	4	(edit)	
	Successful	PC-9	PC-3	ICMP		0.000	N	5	(edit)	

<<Ping between Employees (VLAN 50)>>

PDU List Window

Fire	Last Status	Source	Destination	Type	Color	Time(se	Periodic	Num	Edit	Delete
	Successful	PC-4	PC-2	ICMP		0.000	N	0	(edit)	
	Successful	PC-4	PC-6	ICMP		0.000	N	1	(edit)	
	Successful	PC-4	PC-8	ICMP		0.000	N	2	(edit)	
	Successful	PC-2	PC-8	ICMP		0.000	N	3	(edit)	
	Successful	PC-2	PC-6	ICMP		0.000	N	4	(edit)	
	Successful	PC-6	PC-8	ICMP		0.000	N	5	(edit)	

<<Ping between Executive (VLAN 40)>>

Fire	Last Status	Source	Destination	Type	Color	Time(se	Periodic	Num	Edit	Delete
	Successful	Lapt...	Laptop-1	ICMP		0.000	N	0	(edit)	
	Successful	Lapt...	Laptop-3	ICMP		0.000	N	1	(edit)	
	Successful	Lapt...	Laptop-4	ICMP		0.000	N	2	(edit)	
	Successful	Lapt...	Laptop-4	ICMP		0.000	N	3	(edit)	
	Successful	Lapt...	Laptop-1	ICMP		0.000	N	4	(edit)	
	Successful	Lapt...	Laptop-4	ICMP		0.000	N	5	(edit)	

<<Ping between Finance (VLAN 30)>>

Fire	Last Status	Source	Destination	Type	Color	Time(se	Periodic	Num	Edit	Delete
	Failed	PC-5	PC-4	ICMP		0.000	N	0	(edit)	
	Failed	PC-5	PC-6	ICMP		0.000	N	1	(edit)	
	Failed	PC-5	PC-8	ICMP		0.000	N	2	(edit)	
	Failed	PC-5	PC-2	ICMP		0.000	N	3	(edit)	
	Failed	PC-7	PC-6	ICMP		0.000	N	4	(edit)	
	Failed	PC-7	PC-4	ICMP		0.000	N	5	(edit)	
	Failed	PC-7	PC-8	ICMP		0.000	N	6	(edit)	
	Failed	PC-7	PC-2	ICMP		0.000	N	7	(edit)	
	Failed	PC-9	PC-8	ICMP		0.000	N	8	(edit)	
	Failed	PC-9	PC-2	ICMP		0.000	N	9	(edit)	
	Failed	PC-9	PC-6	ICMP		0.000	N	10	(edit)	
	Failed	PC-9	PC-4	ICMP		0.000	N	11	(edit)	
	Failed	PC-3	PC-2	ICMP		0.000	N	12	(edit)	
	Failed	PC-3	PC-4	ICMP		0.000	N	13	(edit)	
	Failed	PC-3	PC-6	ICMP		0.000	N	14	(edit)	
	Failed	PC-3	PC-8	ICMP		0.000	N	15	(edit)	

<<Ping between Employees (VLAN 50) and Executive (VLAN 40)>>

Fire	Last Status	Source	Destination	Type	Color	Time(se	Periodic	Num	Edit	Delete
	Failed	PC-5	Laptop-2	ICMP		0.000	N	0	(edit)	
	Failed	PC-5	Laptop-3	ICMP		0.000	N	1	(edit)	
	Failed	PC-5	Laptop-1	ICMP		0.000	N	2	(edit)	
	Failed	PC-5	Laptop-4	ICMP		0.000	N	3	(edit)	
	Failed	PC-7	Laptop-3	ICMP		0.000	N	4	(edit)	
	Failed	PC-7	Laptop-4	ICMP		0.000	N	5	(edit)	
	Failed	PC-7	Laptop-1	ICMP		0.000	N	6	(edit)	
	Failed	PC-7	Laptop-2	ICMP		0.000	N	7	(edit)	
	Failed	PC-9	Laptop-4	ICMP		0.000	N	8	(edit)	
	Failed	PC-9	Laptop-3	ICMP		0.000	N	9	(edit)	
	Failed	PC-9	Laptop-1	ICMP		0.000	N	10	(edit)	
	Failed	PC-9	Laptop-2	ICMP		0.000	N	11	(edit)	
	Failed	PC-3	Laptop-1	ICMP		0.000	N	12	(edit)	
	Failed	PC-3	Laptop-2	ICMP		0.000	N	13	(edit)	
	Failed	PC-3	Laptop-4	ICMP		0.000	N	14	(edit)	
	Failed	PC-3	Laptop-3	ICMP		0.000	N	15	(edit)	

<<Ping between Employees (VLAN 50) and Finance (VLAN 30)>>

Fire	Last Status	Source	Destination	Type	Color	Time(se	Periodic	Num	Edit	Delete
	Successful	PC-4	Laptop-2	ICMP		0.000	N	0	(edit)	
	Successful	PC-4	Laptop-3	ICMP		0.000	N	1	(edit)	
	Successful	PC-4	Laptop-1	ICMP		0.000	N	2	(edit)	
	Successful	PC-4	Laptop-4	ICMP		0.000	N	3	(edit)	
	Successful	PC-6	Laptop-3	ICMP		0.000	N	4	(edit)	
	Successful	PC-6	Laptop-4	ICMP		0.000	N	5	(edit)	
	Successful	PC-6	Laptop-1	ICMP		0.000	N	6	(edit)	
	Successful	PC-6	Laptop-2	ICMP		0.000	N	7	(edit)	
	Successful	PC-8	Laptop-4	ICMP		0.000	N	8	(edit)	
	Successful	PC-8	Laptop-1	ICMP		0.000	N	9	(edit)	
	Successful	PC-8	Laptop-3	ICMP		0.000	N	10	(edit)	
	Successful	PC-8	Laptop-2	ICMP		0.000	N	11	(edit)	
	Successful	PC-2	Laptop-1	ICMP		0.000	N	12	(edit)	
	Successful	PC-2	Laptop-2	ICMP		0.000	N	13	(edit)	
	Successful	PC-2	Laptop-4	ICMP		0.000	N	14	(edit)	
	Successful	PC-2	Laptop-3	ICMP		0.000	N	15	(edit)	

<<Ping between Finance (VLAN 30) and Executive (VLAN 40)>>

PC-4

Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 8.8.8.8

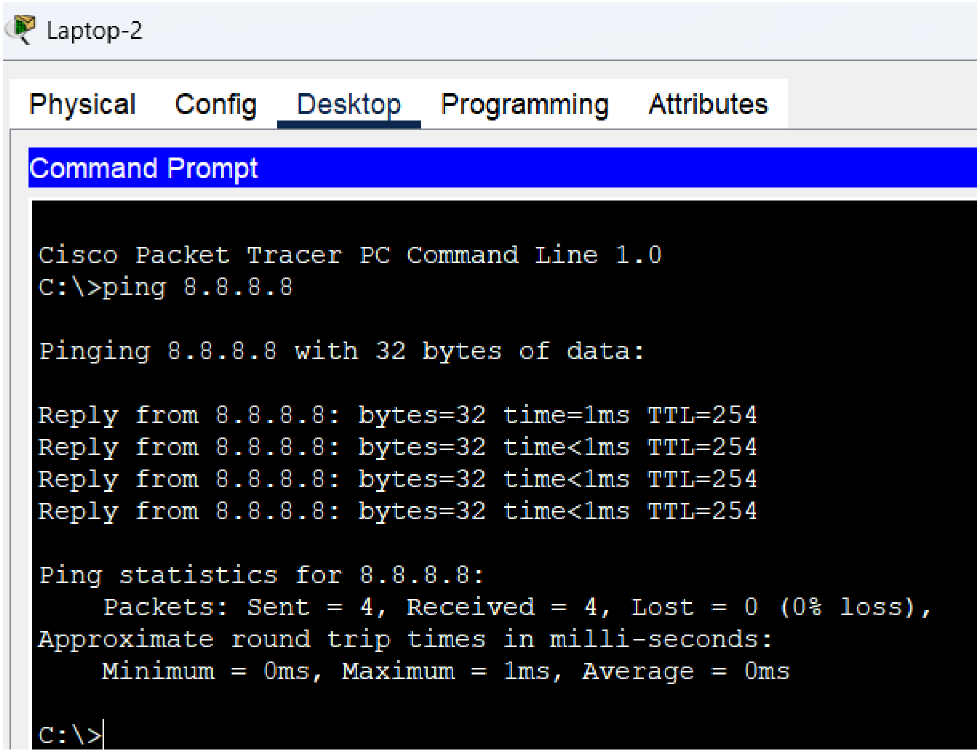
Pinging 8.8.8.8 with 32 bytes of data:

Request timed out.
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254
Reply from 8.8.8.8: bytes=32 time<1ms TTL=254
Reply from 8.8.8.8: bytes=32 time=1ms TTL=254

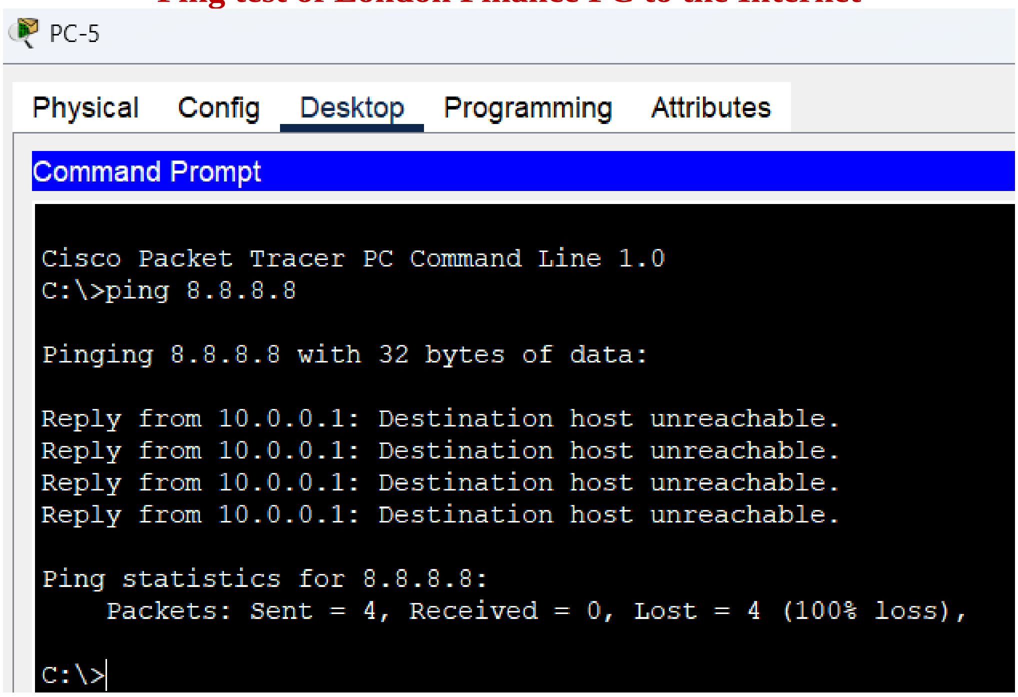
Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

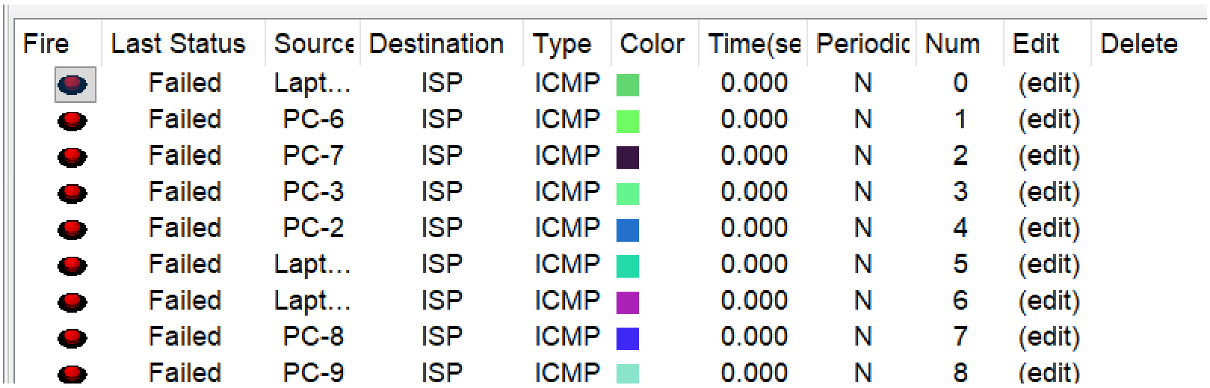
<<Ping test of London Executive PC to the Internet>>

- 

<<Ping test of London Finance PC to the Internet>>

- 

<<Ping test of London Employees PC to the Internet >>
The test failed due to ACL restrictions

- 

Fire	Last Status	Source	Destination	Type	Color	Time(se	Periodic	Num	Edit	Delete
	Failed	Lapt...	ISP	ICMP		0.000	N	0	(edit)	
	Failed	PC-6	ISP	ICMP		0.000	N	1	(edit)	
	Failed	PC-7	ISP	ICMP		0.000	N	2	(edit)	
	Failed	PC-3	ISP	ICMP		0.000	N	3	(edit)	
	Failed	PC-2	ISP	ICMP		0.000	N	4	(edit)	
	Failed	Lapt...	ISP	ICMP		0.000	N	5	(edit)	
	Failed	Lapt...	ISP	ICMP		0.000	N	6	(edit)	
	Failed	PC-8	ISP	ICMP		0.000	N	7	(edit)	
	Failed	PC-9	ISP	ICMP		0.000	N	8	(edit)	

<<All PC pings to ISP except London >>

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7. Conclusion & Recommendations

This network design effectively addresses the requirements for scalability, security, and efficiency. The implementation of OSPF routing, VLAN segmentation, ACL enforcement, and DHCP-based IP management ensures a resilient and future-proof network. Further improvements may include cloud-based security solutions and zero-trust architecture.

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8. References

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- (1997) *RFC 2131: Dynamic Host Configuration Protocol*.

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9. Appendix

Packet Tracer Access Credentials:
Priveleged EXEC mode password: *admin123*
Console/VTY Password: *admin123*